

Air Quality Analysis Report: Del Norte, New Mexico

Comprehensive Assessment of Pollutant Measurements and Environmental Patterns

Executive Summary

This analysis examines air quality data collected at the Del Norte monitoring station (35.14°N, 106.58°W) in New Mexico from December 1, 2025, through January 1, 2026. The dataset comprises 4,284 individual measurements across six regulated air pollutants, revealing significant variability in particulate matter concentrations and consistently low levels of gaseous pollutants. The monitoring period captured a critical air quality episode in mid-December, characterized by extreme PM10 concentrations that warrant immediate attention from environmental and public health officials.

Additional correlation analysis revealed that higher PM10 pollution categories were consistently associated with increased PM2.5, CO, and NO2 concentrations. The analysis also identified a recurring daily pollution cycle, with particulate pollution levels peaking during afternoon and early evening hours. Conversely, higher ozone (O3) categories were generally associated with lower particulate matter concentrations, suggesting an inverse relationship between ozone levels and particulate pollution during the monitoring period.

Primary Pollutants and Measurement Frequency

Pollutant Inventory

The monitoring station tracks six major air pollutants distributed across two measurement categories:

Particulate Matter (Highest Priority):

- PM10 (Coarse Particles): 737 measurements (17.2% of total dataset)
- PM2.5 (Fine Particles): 737 measurements (17.2% of total dataset)

Gaseous Pollutants:

- Ozone (O3): 728 measurements (17.0%)
- Carbon Monoxide (CO): 727 measurements (17.0%)
- Nitrogen Dioxide (NO2): 666 measurements (15.6%)
- Sulfur Dioxide (SO2): 689 measurements (16.1%)

The relatively balanced distribution across pollutants reflects a comprehensive monitoring strategy with consistent sampling protocols throughout the monitoring period.

Pollutant Concentration Overview

Pollutant	Units	Average	Range	Peak Value
PM10	µg/m³	19.49	2–106	106
PM2.5	µg/m³	5.74	0.8–39.9	39.9
CO	ppm	0.26	0–1.1	1.1
NO2	ppm	0.03	0–0.13	0.13
O3	ppm	0.02	0–0.05	0.05
SO2	ppm	0.00	0–0	0

Notable Finding:

Particulate matter measurements demonstrate substantially greater concentration and variability compared to gaseous pollutants, indicating that particulate sources represent the primary air quality concern for this region.

Temporal Patterns and Air Quality Trends

Monthly Progression

The monitoring period revealed distinct phases in air quality conditions:

Phase 1: Baseline Period (December 1–9)

- PM10 averaged 12–17.5 µg/m³
- Relatively stable air quality with low pollutant concentrations
- Minimal day-to-day variation in measurements

Phase 2: Deterioration Phase (December 10–16)

- Progressive increase in PM10 concentrations
- Daily averages rising from 9.96 $\mu\text{g}/\text{m}^3$ to 23.67 $\mu\text{g}/\text{m}^3$
- First indicator of approaching air quality episode

Phase 3: Critical Episode (December 17–19)

- Severe air quality degradation
- PM10 concentrations reaching 95–106 $\mu\text{g}/\text{m}^3$
- Multiple readings in the 80+ $\mu\text{g}/\text{m}^3$ range indicating hazardous conditions
- Absolute peak: 106 $\mu\text{g}/\text{m}^3$ recorded December 18 at 1:00 PM

Phase 4: Secondary Peak (December 20–26)

- PM10 levels remain elevated (20–51 $\mu\text{g}/\text{m}^3$)
- Continued atmospheric instability with fluctuating daily averages
- Secondary peak on December 31: 44.13 $\mu\text{g}/\text{m}^3$ average with individual readings reaching 71 $\mu\text{g}/\text{m}^3$

Phase 5: Stabilization (December 27–January 1)

- Gradual decline in overall pollution levels
 - Return toward baseline concentrations observed in early December
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Diurnal (Daily) Cycle Pattern

Analysis of hourly measurements reveals a pronounced and consistent daily pattern in pollution levels:

Low Pollution Hours (11:00 PM – 11:00 AM)

- PM10 averages: 12–15 $\mu\text{g}/\text{m}^3$
- Cleanest air quality between 11:00 PM and noon
- Reflects overnight atmospheric mixing and reduced emission sources

Transition Hours (11:00 AM – 2:00 PM)

- Gradual increase in concentrations begins
- PM10 rising from 12 to 16 $\mu\text{g}/\text{m}^3$
- Associated with increased daytime activity and changing atmospheric dynamics

Peak Pollution Period (3:00 PM – 5:00 PM)

- Highest average concentrations of the day

- PM10 averages: 23–33 $\mu\text{g}/\text{m}^3$ (peak at 5:00 PM: 32.97 $\mu\text{g}/\text{m}^3$)
- PM2.5 follows similar pattern, peaking around 5:00 PM
- Represents 2–2.5 $\times$ increase from overnight baseline

Decline Phase (6:00 PM – 11:00 PM)

- Gradual reduction in pollutant levels
- Return toward diurnal minimum by midnight

Interpretation:

The consistent 3:00–5:00 PM pollution peak suggests multiple contributing factors:

- Rush hour traffic emissions
 - Atmospheric boundary layer contraction
 - Cumulative daytime emissions concentrating in reduced air volume
 - Possible household heating and cooking activities in early winter months
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Correlation Analysis and Pollutant Relationships

Additional analysis was conducted to evaluate relationships between different air pollutants and identify how pollutant concentrations changed under varying air quality conditions.

PM10 Category Relationships

Pollution categories were grouped into Low, Moderate, High, and Very High PM10 concentration ranges. The analysis revealed a consistent increase in PM2.5, CO, and NO₂ concentrations as PM10 levels increased:

Low PM10 Category

- PM2.5 average: 3.61 $\mu\text{g}/\text{m}^3$
- CO average: 0.183 ppm
- NO₂ average: 0.018 ppm

Moderate PM10 Category

- PM2.5 average: 8.24 $\mu\text{g}/\text{m}^3$
- CO average: 0.363 ppm
- NO₂ average: 0.037 ppm

High PM10 Category

- PM2.5 average: 12.02 $\mu\text{g}/\text{m}^3$
- CO average: 0.475 ppm
- NO2 average: 0.049 ppm

Very High PM10 Category

- PM2.5 average: 14.24 $\mu\text{g}/\text{m}^3$
- CO average: 0.530 ppm
- NO2 average: 0.058 ppm

These findings suggest that particulate matter pollution events were commonly accompanied by elevated combustion-related pollutants, indicating shared contributing sources such as traffic emissions, household heating, or atmospheric stagnation conditions.

Ozone Relationship Patterns

The analysis also revealed that higher ozone (O3) categories were generally associated with lower PM10 and PM2.5 concentrations:

Low O3 Category

- Average PM10: 28.12 $\mu\text{g}/\text{m}^3$
- Average PM2.5: 9.32 $\mu\text{g}/\text{m}^3$

Moderate O3 Category

- Average PM10: 15.63 $\mu\text{g}/\text{m}^3$
- Average PM2.5: 4.41 $\mu\text{g}/\text{m}^3$

High O3 Category

- Average PM10: 13.00 $\mu\text{g}/\text{m}^3$
- Average PM2.5: 2.91 $\mu\text{g}/\text{m}^3$

This inverse relationship may reflect differing atmospheric conditions influencing ozone formation versus particulate matter accumulation.

PM2.5-to-PM10 Ratio Trends

The PM2.5-to-PM10 ratio declined as PM10 pollution categories increased:

- Low PM10 category: 0.36
- Moderate PM10 category: 0.305

- High PM10 category: 0.247
- Very High PM10 category: 0.206

This pattern suggests that coarse particulate matter became increasingly dominant during severe pollution episodes, potentially indicating stronger contributions from dust, mechanical particles, or atmospheric resuspension sources during high-pollution periods.

Extreme Pollution Events and Peak Concentrations

Critical Air Quality Episode: December 17–19, 2025

Six extreme pollution events ($\text{PM}_{10} \geq 80 \text{ }\mu\text{g}/\text{m}^3$) occurred during a concentrated 72-hour period:

Date	Day of Week	Time	PM10 ($\mu\text{g}/\text{m}^3$)	Pollution Level
December 17	Wednesday	5:00 PM	95	Severe
December 17	Wednesday	6:00 PM	94	Severe
December 18	Thursday	1:00 PM	106	Hazardous
December 18	Thursday	2:00 PM	87	Severe
December 19	Friday	6:00 PM	94	Severe
December 22	Monday	5:00 PM	87	Severe

Peak Concentration Analysis

The maximum recorded PM10 concentration of $106 \text{ }\mu\text{g}/\text{m}^3$ significantly exceeds established air quality standards:

- U.S. EPA 24-hour PM10 Standard: $150 \text{ }\mu\text{g}/\text{m}^3$ (not exceeded)
- Unhealthy Range (AQI): $155\text{--}254 \text{ }\mu\text{g}/\text{m}^3$ (approached but not reached)
- Occupational Safety Standard (OSHA): $5,000 \text{ }\mu\text{g}/\text{m}^3$ (vastly below)
- WHO Air Quality Guideline (24-hour): $45 \text{ }\mu\text{g}/\text{m}^3$ (exceeded by 136%)

Critical Observation:

While the absolute maximum did not exceed the EPA 24-hour standard, the concentration reached levels approaching the “unhealthy” air quality category for sensitive groups. The peak occurred during afternoon hours when outdoor activity is highest, increasing public exposure risk.

Secondary Peak Event: December 31, 2025

A secondary air quality episode occurred on New Year's Eve:

- PM10 daily average: 44.13 $\mu\text{g}/\text{m}^3$
- Individual readings reaching 71 $\mu\text{g}/\text{m}^3$
- PM2.5 daily average: 22.53 $\mu\text{g}/\text{m}^3$
- PM2.5 peak: 39.9 $\mu\text{g}/\text{m}^3$ (approximately 8× higher than typical daily maximum)

Hypothesis:

The December 31 spike may correlate with increased household heating demand during winter weather, combined with New Year's Eve celebrations involving fireworks and increased combustion activities.

Comprehensive Air Quality Summary

Statistical Summary Table

Pollutant	Units	Total Measurements	Average	Minimum	Maximum	Median	25th Percentile	75th Percentile	Std Deviation
PM10	$\mu\text{g}/\text{m}^3$	737	19.49	2.00	106.00	15.00	9.00	24.00	15.83
PM2.5	$\mu\text{g}/\text{m}^3$	737	5.74	0.80	39.90	4.30	2.80	7.30	4.56
O3	ppm	728	0.02	0.00	0.05	0.02	0.01	0.03	0.01
NO2	ppm	666	0.03	0.00	0.13	0.02	0.01	0.04	0.02
CO	ppm	727	0.26	0.00	1.10	0.20	0.10	0.40	0.20
SO2	ppm	689	0.00	0.00	0.00	0.00	0.00	0.00	0.00

Pollutant-Specific Assessment

PM10 (Coarse Particulate Matter)

- Demonstrates the greatest variability and highest absolute concentrations
- Standard deviation of 15.83 $\mu\text{g}/\text{m}^3$ indicates substantial fluctuation

- 75th percentile of 24 $\mu\text{g}/\text{m}^3$ remains below health concern thresholds for general population
- However, extreme values show atmospheric conditions capable of reaching hazardous levels
- Primary pollution concern requiring management attention

PM2.5 (Fine Particulate Matter)

- More stable than PM10 with lower average concentration (5.74 vs. 19.49 $\mu\text{g}/\text{m}^3$)
- Peak concentration of 39.9 $\mu\text{g}/\text{m}^3$ approaches EPA 24-hour standard of 35.4 $\mu\text{g}/\text{m}^3$ (AQI moderate)
- Standard deviation of 4.56 indicates more consistent baseline with occasional spikes
- Secondary concern but still requiring monitoring

Gaseous Pollutants (O3, NO2, CO)

- All remain consistently low throughout monitoring period
- Average concentrations well below established health standards
- Minimal variability except for CO with occasional events up to 1.1 ppm
- Do not require immediate intervention but warrant continued monitoring

Sulfur Dioxide (SO2)

- Zero detection across all 689 measurements
- Indicates absence of SO2-emitting sources (fossil fuel power generation, heavy industry)
- Reflects limited industrial activity in monitoring area

Key Findings and Observations

1. Particulate Matter Dominance

The primary air quality challenge is particulate matter, particularly PM10, not gaseous pollutants. This pattern suggests sources related to dust, mechanical processes, and combustion rather than ozone precursors or vehicle emissions alone.

2. Predictable Daily Variation

The consistent 3:00–5:00 PM pollution peak provides a predictable pattern that could be leveraged for public health advisories and activity recommendations for sensitive populations.

3. Critical Mid-December Episode

The December 17–19 air quality crisis represents an extraordinary event potentially attributable to:

- Atmospheric temperature inversion trapping pollutants
- Regional weather patterns reducing air mixing
- Possible wildfire smoke transport from distant sources
- Peak winter heating demand coinciding with atmospheric stagnation

4. Single Monitoring Location Limitation

Data represents conditions at only one fixed location (Del Norte). Spatial variability across the region remains unknown, potentially affecting public health assessment accuracy.

5. Seasonal Winter Pattern

The concentration of extreme pollution events in mid-to-late December suggests winter-specific factors such as residential heating demand and atmospheric stability patterns characteristic of winter months.

6. Correlated Pollutant Behavior

Correlation analysis demonstrated that elevated PM₁₀ pollution levels were consistently associated with increases in PM_{2.5}, CO, and NO₂ concentrations. This suggests that particulate pollution events were influenced by multiple interconnected emission sources and atmospheric conditions rather than isolated pollutant activity alone.

Recommendations for Environmental and Public Health Officials

Immediate Actions

- Establish enhanced monitoring during December–January winter months
- Develop and communicate air quality advisories for sensitive populations during afternoon hours (2:00–6:00 PM)
- Investigate root causes of the December 17–19 pollution episode for future prevention

Medium-Term Strategies

- Implement public awareness campaigns targeting pollution peak hours
- Coordinate with community organizations to promote indoor activities for sensitive groups during high-pollution periods
- Evaluate emission control measures for identified pollution sources

Long-Term Monitoring

- Establish additional monitoring stations to characterize spatial air quality patterns
- Conduct source apportionment studies to identify primary PM10 contributors
- Develop predictive models for pollution episodes enabling advance warning systems

Conclusion

The Del Norte air quality monitoring data reveals a region with generally acceptable air quality interrupted by significant episodic pollution events concentrated in late December. While gaseous pollutants remain consistently low, particulate matter—particularly PM10—represents the primary environmental and public health concern. The dramatic December 17–19 air quality crisis, coupled with predictable daily pollution peaks, indicates the need for targeted monitoring, public health interventions, and source investigation.

Correlation analysis further strengthened the findings by demonstrating that severe PM10 pollution episodes were closely associated with increases in PM2.5, CO, and NO2 concentrations. The observed inverse relationship between ozone and particulate matter also highlights the complexity of atmospheric pollution behavior and the importance of multi-pollutant monitoring strategies.

Continued data collection and enhanced monitoring strategies will enable more effective environmental protection and public health protection for the Del Norte community.